SUBJECT: DIGITAL SYSTEM DESIGN(DSD)

Sem: 3, EXTC

Question Bank for IA-2

1. Demultiplexer is also called as
   a) Data Distributor
   b) Data Selector
   c) Mux
   d) Input to output coder

2. 1-bit magnitude comparator has
   a) One input and two outputs
   b) Two inputs and two outputs
   c) Two inputs and three outputs
   d) Three inputs and three outputs

3. Carry look ahead logic uses the concept of _____
   a) Inverting the inputs
   b) Complementing the outputs
   c) Ripple factor
   d) Generating and propagating carries

4. The number of 4:1 MUXES required to construct 32:1 MUX are
   a) 11
   b) 8
   c) 10
   d) 12

5. Comparators are used in_____
   a) CPU
   b) Motherboard
   c) Hard Drive
   d) Memory

6. If Strobe input is high then the multiplexer is _____
   a) Enable
   b) Disable
   c) Saturation
   d) High Impedance

7. Demultiplexer is basically a
   a) Parallel to Serial converter
   b) Storage device
   c) Code Converter
   d) Serial to Parallel converter

8. If we record any music in any recorder, such type of process is called _____
   a) Multiplexing
   b) Encoding
   c) Decoding
   d) Demultiplexing

9. A combinational Circuit that changes Code into set of signals is called
   a) Encoder
   b) Code converter
   c) Decoder
   d) Demultiplexer

10. In a BCD adder if the sum is a valid BCD number then the logic state of correction logic is
   a) 0000
   b) 0110
   c) 0111
   d) 1000

11. The Forbidden state of SR Flip Flop is overcomed by
   a) JK master Slave Flip Flop
   b) JK Flip Flop
   c) D Flip Flop
   d) T Flip Flop

12. If the data bit at the input is required to be passed on to output without any change the mostsuitable flip flop is
    a. T Flip Flop
    b. D Flip Flop
    c. SR Flip Flop
    d. JK Flip Flop

13. How is a J-K flip-flop made to

    toggle?a) J = 0, K = 0
    b) J = 1, K = 0
    c) J = 0, K = 1
    d) J = 1, K = 1

14.In a master-slave flip-flop, when is the Slave enabled?
    a. When the Clock is High
    b. When the Clock is Low
    c. During level of the Clock
    d. When the Clock is Disabled

15.What is the significance of the J and K terminals on the J-K flip-flop?

    a. There is no known significance in their designations.
    b. The J represents "jump," which is how the Q output reacts whenever the clock goes highand the J input is also HIGH.
    c. The letters were chosen in honor of Jack Kilby, the inventory of the integrated circuit.
    d. All of the other letters of the alphabet are already in use.

16. A positive edge-triggered D flip-flop will store a 1 when_____.
    a) the D input is HIGH and the clock transitions from HIGH to LOW
    b) the D input is HIGH and the clock transitions from LOW to HIGH
    c) the D input is HIGH and the clock is LOW
    d) the D input is HIGH and the clock is HIGH

17.If both inputs of an S-R flip-flop are LOW, what will happen when the clock goes high?
    a. The output will reset
    b. An invalid state will exist.
    c. The output will toggle.
    d. No change will occur in the output

18. The simplest type of shift register is

a. SISO
b. SIPO
c. PIPO
d. PISO

19. The bit sequence 10011100 is serially entered (right-most bit first) into an 8-bit parallel out shift register that is initially clear. What are the Q outputs after four clock pulses?

a) 10011100
b) 11000000
c) 00001100
d) 11110000

20. Asynchronous counter is also called as

a. Parallel Counter
b. Ripple Counter
c. BCD Counter
d. Mod Counter

21. In Asynchronous Counter using T Flip flop inputs T of all the flip flops is connected to
a) Logic 0
b) Clear Input
c) Clock Input
d) Logic 1

22. Parallel loading of data in a counter refers to
a) Each FF is loaded with data on a separate clock.
b) All FFs are preset with data
c) Counter is Cleared
d) Counter is Stored with all 1's

23. Speed of ripple counter is limited by the propagation delay of
a) Flip flops only with gates
b) Only circuit gates
c) each flip-flop
d) All flip flops and gates

24. Counter which counts randomly without going in sequence is called as____
a) Random Counter
b) Up/down Counter
c) Skipping state Counter
d) Serial Counter

25. State diagram provides
a) exactly the same information as the state table
b) Information about characteristic equation
c) graphical representation of states
d) Information about the truth table

26. Which of the following is not a shift Register Counter
a) Ring Counter
b) BCD Counter
c) Jhonson's Counter
d) Twisted Ring Counter

27. The counter Obtained by connecting complement of output of last FF to the input of the first FF is
a) Jhonson's Counter
b) BCD Counter
c) Ring Counter
d) Up/down Counter

28. Another name for Up Down Counter is

a) Dual counter

b) Multi counter

c) Two counter

d) Multimode Counter

29. The major disadvantage of shift register counters is

a) Not all the states are utilized

b) Complexity in Operation

c) Slow operation

d) Lower Throughput

30. The PLD in which both OR and AND array are programmable is
    a. PLA
    b. PAL
    c. RAM
    d. Buffer

31. Current sourcing capability of PLA can be improved by using _____
    a. Input Buffer
    b. Memory Units
    c. Output Buffer
    d. Invert/Non-Invert Matrix

32. The major disadvantage of RAM is
    a. Access speed is too slow
    b. Its matrix size is too big
    c. High power consumption
    d. It is volatile

33. The difference between static RAM and dynamic RAM is
    a. Static RAM must be refreshed, dynamic RAM does not
    b. There is no difference
    c. Dynamic RAM must be refreshed, static RAM does not
    d. SRAM is slower than DRAM

34. The inputs in the PLD is given through _____

a. NAND gates
b. OR gates
c. AND gates
d. NOR gates

35. Which of the following is electrically erased and reprogrammed by the user?

a. EEPROM
b. PROM
c. RAM
d. Flip Flop

36. Boolean functions in standard SOP form only can be implemented using

a. PAL
b. PLA
c. RAM
d. PROM

37. PAL is an abbreviation of

a. Programmable AND Logic
b. Programmable Array Logic
c. Preset Array Logic
d. Preset AND Logic

38. Language that describes the behaviour or structure of digital circuit /IC's is

a. Hardware Data Language
b. Hardware Description Language
c. Hardware Description Logic
d. Hard Description Language

39. VHDL Program format consists

a. Library declaration,Architecture
b. Library declaration,Entity
c. Entity, Architecture
d. Library declaration,Entity,Architecture

40. There are_____types of Port Modes

a. 3
b. 2
c. 4
d. 5

41. Keyword Process is used in_____modeling

a. Data Flow
b. Structural
c. Mixed
d. Behavioural

42. Boolean expressions are used in_____modeling

a. Data Flow
b. Structural
c. Mixed
d. Behavioural

43. The number of OR gates required for a Decimal-to-BCD encoder are

a. 2
b. 10
c. 4
d. 3

44. Which of the following is INCORRECT with reference to PLA

a. It has both programmable OR and AND array
b. It provides full decoding to the variables
c. One application of a PLA is to implement the control over a datapath
d. Massive functions can be implemented uing PLA

45. The minimum number of inputs required to be specified in the entity with which program for UP/DOWN counter can be written in VHDL are

a. 3
b. 2
c. 5
d. 6

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